Prodigy Anesthesia Quick Review

Intercostal Nerve Block

What are the indications for performing an intercostal nerve block?	Intercostal nerve blocks are not typically performed as the sole anesthetic for surgical procedures, but are instead used for postoperative pain management for thoracic or upper abdominal procedures or for pain due to shingles, cancer, or rib fractures. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1018.*
How is an intercostal nerve block performed?	The patient is positioned in the supine or lateral decubitus position. The correct rib is identified by palpation and then a 22 or 25 gauge needle is walked off the inferior border of the rib and advanced about 0.5 cm. After careful aspiration for blood or air, 3 to 5 mL of local anesthetic is injected. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1018.*
What are the risks of intercostal nerve block?	Because of the proximity of the nerve to the vascular structures in the intercostal neurovascular bundle, intravascular injection is a risk. Because of the proximity of the needle tip to the pleura, it is possible to cause a pneumothorax. Also, an intercostal injection results in the highest blood levels of local anesthetic per volume injected for block of a specific nerve, so drug toxicity is a potential risk if large amounts of local anesthetic are injected. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1018.*
How are the structures of the intercostal neurovascular bundle positioned?	The intercostal neurovascular bundle travels underneath the rib. From superior to inferior lie the vein, artery, and nerve. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1018.*
From where do the intercostal nerves originate?	The intercostal nerves originate from the dorsal and ventral rami of the thoracic spinal nerves. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1018.*
Along what path do the intercostal nerves travel?	After originating from the thoracic spinal nerves, the intercostal nerves exit at the intervertebral foramina and travel along the underside of the rib. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1018.*

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